

SECTION 014100 – QUALITY REQUIREMENTS

PART 1 GENERAL

1.1 PURPOSE

- A. The Subcontractor is responsible for all activities necessary to manage, control, and document work so as to ensure compliance with the Subcontract documents. The Subcontractor's responsibility includes ensuring adequate Quality Control services are provided for work accomplished on- and off-site by their organization, suppliers, sub-tier contractors, technical laboratories, and consultants.
- B. The work activities include safety, submittal management, testing and inspection, and all other functions relating to the requirement for quality construction. The Subcontractor shall implement a Quality Control program and designate a quality responsible person(s) to achieve these goals. Depending on the size and complexity of the project, a dedicated Quality Control Project Manager may be required.

1.2 REFERENCES

- A. The publications listed below form a part of this subcontract to the extent referenced.
 - 1. 010010 General Requirements
 - 2. 013100 Environment, Safety, and Health Requirements
 - 3. 013300 Submittals

1.3 THE QUALITY CONTROL PROGRAM

- A. Corporate Quality Assurance Plan: The Subcontractor shall furnish with the bid or proposal a copy of their corporate Quality Assurance Plan. Failure to submit such a document could constitute a non-responsive bid or proposal. See Appendix A for Corporate Quality Assurance Plan guidance. If the Subcontractor does not have a Corporate Quality Assurance Plan developed at time of bid, they may elect to use form [Appendix C] to satisfy this requirement.
- B. Project Quality Control Plan: The Subcontractor/Vendor shall furnish a Project Quality Control (PQC) Plan to the Procurement Administrator, for review by FermiForward, not later than 10 working days after receipt of award of Subcontract.

The PQC Plan shall identify personnel, procedures, controls, instructions, test, records, and forms to be used specific to this project at Fermilab. The PQC Plan shall be updated to incorporate any changes in scope of work, due to directed field changes or supplemental agreements, as these changes occur. See Appendix B for Project Quality Control Plan guidance.

1.4 REQUIREMENTS

- A. The PQC Plan shall include, as a minimum, the following information to cover all construction operations, both onsite and offsite, including work by sub-tier contractors, fabricators, suppliers, and purchasing agents:
1. The name, qualifications, duties, responsibilities, and authorities of each person assigned a Quality Control function, including the identity of the Project Quality Control Manager:
 2. Depending on the size and complexity of the project, the project superintendent and/or competent person may also serve as the Quality Control Project Manager.
- B. As a minimum, the Project Quality Control Manager shall be responsible for the following:
1. Ensure the defined Project Quality Control Plan is effectively implemented.
 2. Interface with the Construction Management Office (CMO) on all quality matters;
 3. Maintain all quality related records;
 4. Review / submit shop drawings;
 5. Participate in all preparatory, initial and follow-up meetings;
 6. Attend weekly construction meetings;
 7. Assure all required testing is performed;
 8. Develop and maintain a deficiency/nonconformance list;
 9. Prepare and submit daily quality control reports;
 10. Participate in Authorization for Use and Possessions (AUP), punch list and final inspections.
- C. In addition, the Subcontractor shall provide as part of the Quality Control organization, specialized personnel to assist the Project Quality Control Manager for the discipline(s) specified below. These individuals may be employees of the Subcontractor or Sub-tier contractor; will report to the Project Quality Control Manager; must be physically present at the construction site during work on their areas of responsibility; must have the necessary education and/or experience in Quality Assurance / Quality Control for similar activities. . These individuals may perform other duties but must be allowed sufficient time to perform their assigned Quality Control duties as described **010010**.
1. Control, verification, and acceptance testing procedures for each specific test required by the Subcontractor.
 2. Procedures for tracking preparatory, initial, and follow-up control phases and control, verification, and acceptance tests including documentation.
 3. A list of the definable features of work. A definable feature of work is a task that is separate and distinct from other tasks, has separate control requirements, and may be identified by different trades or disciplines, or it may be work by the same trade in a different environment.

1.5 ACCEPTANCE

- A. Acceptance of the Subcontractor's PQC Plan is required prior to Notice to Proceed. Acceptance is conditional and will be predicated on satisfactory performance during the construction.

- B. FermiForward reserves the right to require the Subcontractor to make modifications to their PQC Plan and operations including personnel as necessary to obtain the quality specified (Construction Subcontract Terms and Conditions). FermiForward review and acceptance will occur within 5 working days of the submittal of the PQC Plan.

PART 2 PRODUCTS - Not Used

PART 3 EXECUTION – Not Used

END OF SECTION

Appendix A – Corporate Quality Assurance (QA) Plan Guidance

The section below describes the elements expected in the Corporate Quality Assurance Plans submitted by Subcontractors.

- **Purpose**
 - The Subcontractor is to describe the purpose of the document (Corporate QA Plan) as it relates to their organization.
- **Roles, Responsibilities, and Authorities**
 - The Subcontractor is to describe the roles, responsibilities, and authorities relating to the various Quality Assurance processes described within this plan.
- **Scope**
 - The Subcontractor is to describe the scope of the Corporate QA Plan.
- **Organizational Chart**
 - The Subcontractor is to highlight the QA roles in the organization.
- **Requirements Verification**
 - The Subcontractor is to describe how requirements relating to the scope of work will be verified.
- **Document Control**
 - The Subcontractor is to describe how documentations is controlled, including Project Drawings. The Request for Information process should also be described.
- **Submittals/Procurement**
 - The Subcontractor is to describe their submittal process, ensuring it complies with project plans and specifications.
- **Material and Equipment Control**
 - The Subcontractor is to describe how materials and equipment are received, stored, maintained, and released.
- **Turnover Records**
 - The Subcontractor is to describe what documentation will be turned over to the Owner and at what point in time.
- **Procedures**
 - The Subcontractor is to describe the various procedures that are used to control the performance of work to ensure requirements for design, materials, fabrication, erection, installation, etc. are met.
- **Three Phases of Control**
 - The Subcontractor is to describe how the three phases of control are carried out (preparatory, initial, and follow-up).
- **Suspect/Counterfeit Items**
 - The Subcontractor is to describe processes that will be in place for monitoring and controlling the use of suspect/counterfeit items.
- **Nonconformance Handling**
 - The Subcontractor is to describe processes that will be in place for managing nonconforming goods or processes.
- **Lessons Learned**
 - The Subcontractor is to describe their lessons learned process, including identifying, documenting, and sharing lessons.
- **Continuous Improvement**

- The Subcontractor is to describe processes that are in place to identify and implement improvement opportunities.
- **Inspection and Testing**
 - The Subcontractor is to describe the inspection and testing requirements and methods for documentation. Additionally, the qualifications of personnel performing inspections and tests should also be described. Also, the managing of testing equipment including calibration requirements should also be described.
- **Definable Features of Work (DFOW)**
 - The Subcontractor is to list each DFOW and describe the following for each:
 - Activities performed prior to, during, and after work is completed
 - Associated Requirements and Documentation
- **Commissioning**
 - The Subcontractor is to describe the commissioning process that is followed.

Appendix B – Project Quality Control Plan Guidance

- **Purpose**
 - Describe the purpose of the document as it relates to the specific project that's been awarded.
- **Project Specific Organization**
 - Describe the personnel that will be working directly on the project, highlighting the quality control responsible individuals and their lines of authority. An organizational chart should be provided as well as the names and qualifications of each member. Describe the duties, responsibilities, and authorities of each person assigned a QC function.
- **List of DFW's for the Project**
 - Provide a list of Definable Features of Work specific to the project and describe the associated procedures for each. All known subcontractors shall be identified with their corresponding DFW
- **QC Procedures/Submittals**
 - Provide the control, verification, and acceptance testing procedures for each test to be performed.
 - Include the test name, feature of work to be tested, testing frequency, and responsible party for each test. This includes QC performed both internally and by independent testing agents.
 - Describe the procedures for scheduling, reviewing, certifying, and managing submittals, including those of subcontractors, designers of record, consultants, architect engineers, offsite fabricators, suppliers, and purchasing agents.
- **QC Tracking/Verification**
 - Describe the procedures for tracking preparatory, initial, and follow-up control phases and control, verification, and acceptance test documentation.
- **Nonconformance Handling**
 - Describe the procedures for tracking deficiencies/nonconformances from identification through corrective action implementation. Establish verification procedures that identified deficiencies/nonconformances have been corrected.
 - Include communication process for deficiencies/nonconformances.
- **Meetings and Reports**
 - The Subcontractor is to describe what meetings will take place, including but not limited to, Construction Preparatory Phase, Construction Initial Phase, and Construction Follow Up Phase meetings. Reports/documentation that is generated at each meeting should also be described.

Appendix C – Corporate Quality Assurance (QA) Plan

Quality Assurance Plan

To be used by contractors who do not submit their own QA Plan

1. Purpose

This Quality Assurance (QA) Plan establishes minimum quality standards, responsibilities, and procedures that the Contractor must follow to ensure that all work meets the contract requirements, applicable codes, and industry best practices. By signing this document the Contractor agrees to adopt and comply with this QA Plan for the duration of the project.

2. Roles, Responsibilities, and Authorities

- Contractor QA Representative

The Contractor shall designate a QA Representative who will be responsible for overseeing quality-related activities, performing inspections, and coordinating with the Owner's Quality personnel.

- Project Manager / Superintendent

Responsible for ensuring that project staff and subcontractors adhere to this QA Plan and perform work according to approved drawings and specifications.

- Foremen / Trade Supervisors

Responsible for daily implementation of quality requirements in their respective work areas.

3. Quality Objectives

- Deliver work that complies with all project specifications, approved drawings, and applicable codes.
- Prevent rework by ensuring quality is built in at every stage.
- Maintain consistent workmanship and construction practices across all trades.

4. Submittals and Approvals

- All required submittals (shop drawings, material certifications, product data, etc.) shall be submitted and approved before work begins.
- Deviations from approved submittals or specifications must be approved in writing by the Owner or their representative.

5. Inspection and Testing

- Pre-Work Inspection

Conduct inspections before work starts to ensure site conditions are suitable and all materials/tools meet specified requirements.

- In-Process Inspections

Work shall be inspected during execution to identify deficiencies early and ensure compliance.

- Final Inspection

All work shall be subject to a final inspection. Any deficiencies must be corrected before acceptance.

- Testing

Testing shall be performed as required by contract documents, applicable codes, or at the direction of the Owner. Testing may include concrete testing, weld inspections, waterproofing tests, etc.

6. Nonconforming Work

- All nonconforming work must be documented, evaluated, and corrected in accordance with project specifications.
- Rework shall be completed at the Contractor's expense and scheduled to minimize project impacts.

7. Documentation and Records

- Daily QA records shall be maintained by the Contractor and made available upon request.
- Records include: inspection checklists, test reports, material certifications, delivery tickets, and nonconformance logs.

8. Corrective and Preventive Action

- When quality issues are discovered, root causes shall be identified, and corrective measures shall be implemented to prevent recurrence.
- Preventive actions should be taken proactively when risks or trends are identified.

9. Training and Communication

- All personnel performing work under this plan shall be trained in quality requirements relevant to their duties.
- QA responsibilities and standards shall be communicated during pre-task meetings or toolbox talks.

10. Project Closeout

- Prior to final acceptance, the Contractor shall complete all punch list items and submit all required closeout documentation including warranties, as-builts, and test reports.
- Final QA review shall be conducted to ensure all deliverables are complete and compliant.

Acknowledgment

By signing below, the Contractor agrees to adopt this Quality Assurance Plan in full and understands that failure to comply may result in rework, payment withholding, or termination for cause.

Contractor Name	Signature	Title	Date